

PSTAT 175: Final Project

Matthew Aydin, Angel Alcantara, Adam Sztonyk

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1 Introduction

The **burn** dataset, originally from Ichida and available in the **KMsurv** package (Klein), contains clinical records of 154 burn patients. This dataset was collected to study factors influencing burn wound infections, particularly *Staphylococcus aureus* infections, in a clinical setting. We define the failure time as **T3**, representing the number of days until *Staphylococcus aureus* infection or censoring, with the event indicator **D3** (1 = infection, 0 = censored). Additionally, we consider the excision event at time **T1** (days to excision or time on study) with indicator **D1** (1 = excision performed, 0 = no excision), which we will model as a time-varying covariate to explore its impact on infection risk.

The dataset includes the following covariates:

- **Z1**: Treatment type (0 = routine bathing, 1 = body cleansing)
- **Z2**: Gender (0 = male, 1 = female)
- **Z3**: Race (0 = nonwhite, 1 = white)
- **Z4**: Percent of total surface area burned (continuous, in percentage units)
- **Z5-Z10**: Binary indicators (0 = no, 1 = yes) for burn sites: head (Z5), buttock (Z6), trunk (Z7), upper leg (Z8), lower leg (Z9), respiratory tract (Z10)
- **Z11**: Type of burn (1 = chemical, 2 = scald, 3 = electric, 4 = flame)

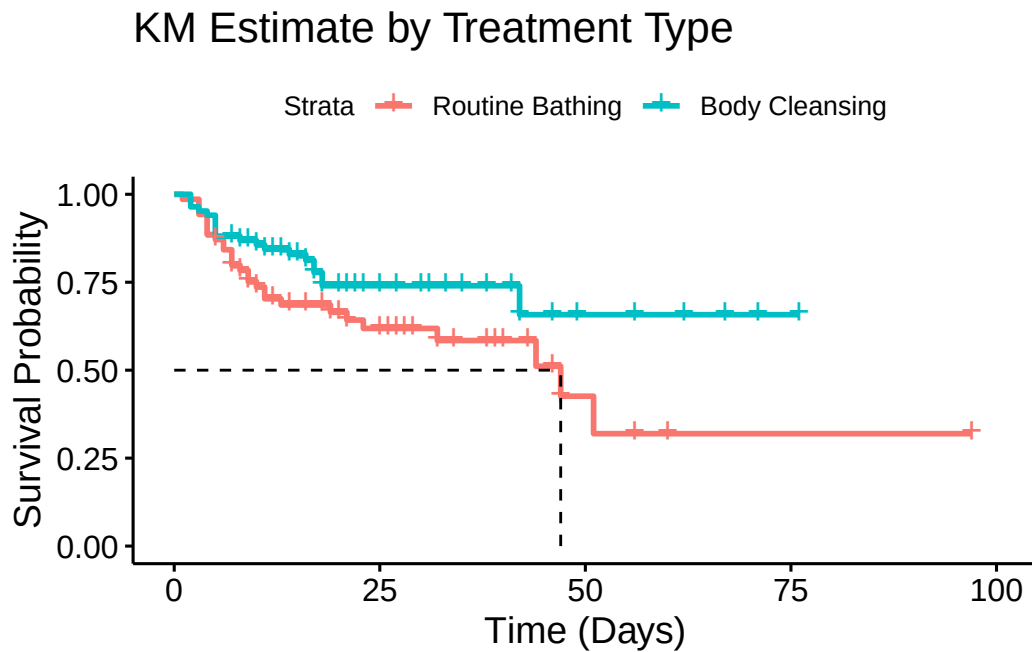
The primary scientific question motivating this study is: *How does the body cleansing treatment (Z1=1) affect the hazard of Staphylococcus aureus infection compared to routine bathing (Z1=0), and does this effect vary with the excision event (T1), while accounting for other patient and burn characteristics?* To address this, we will extend our analysis beyond the standard Cox model by incorporating the time-varying nature of the excision event.

```
burn.surv <- Surv(time = burn$T3, event = burn$D3)
ggsurvplot(
  survfit(burn.surv ~ Z1, data = burn),
  surv.median.line = "hv",
```

```

legend.labs = c("Routine Bathing", "Body Cleansing"),
xlab = "Time (Days)",
ylab = "Survival Probability",
title = "KM Estimate by Treatment Type"
)

```



2 Model Fitting

We begin with Kaplan-Meier estimation, followed by fitting Cox proportional hazards models. We use forward stepwise selection with AIC to identify key covariates, starting with a standard Cox model and later extending it to include time-varying effects.

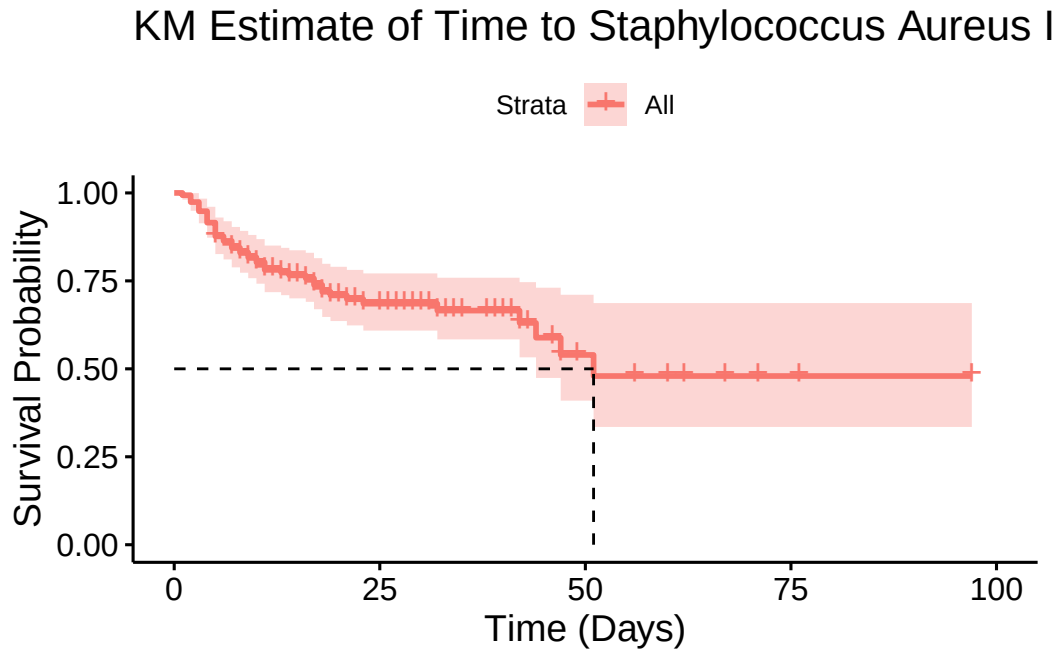
2.1 Kaplan-Meier Estimate

```

ggsurvplot(
  survfit(burn.surv ~ 1),
  surv.median.line = "hv",
  data = burn,
  xlab = "Time (Days)",

```

```
ylab = "Survival Probability",
title = "KM Estimate of Time to Staphylococcus Aureus Infection"
)
```



The overall Kaplan-Meier curve shows a median survival time of approximately 51 days, offering a baseline infection risk estimate before covariate adjustment.

2.2 Cox Proportional Hazards Model

2.2.1 Full Model

```
cox_full <- coxph(burn.surv ~ Z1 + Z2 + Z3 + Z4 + Z5 +
                  Z6 + Z7 + Z8 + Z9 + Z10 + as.factor(Z11), data = burn)
summary(cox_full)
```

Call:

```
coxph(formula = burn.surv ~ Z1 + Z2 + Z3 + Z4 + Z5 + Z6 + Z7 +
      Z8 + Z9 + Z10 + as.factor(Z11), data = burn)
```

n= 154, number of events= 48

	coef	exp(coef)	se(coef)	z	Pr(> z)
Z1	-0.651754	0.521131	0.323330	-2.016	0.0438 *
Z2	-0.556911	0.572976	0.405182	-1.374	0.1693
Z3	2.149127	8.577367	1.040139	2.066	0.0388 *
Z4	0.002041	1.002043	0.009843	0.207	0.8357
Z5	-0.014035	0.986063	0.370920	-0.038	0.9698
Z6	0.541461	1.718516	0.430265	1.258	0.2082
Z7	-0.055650	0.945870	0.507956	-0.110	0.9128
Z8	-0.171817	0.842133	0.393707	-0.436	0.6625
Z9	-0.324566	0.722841	0.373905	-0.868	0.3854
Z10	0.228682	1.256943	0.372930	0.613	0.5397
as.factor(Z11)2	1.527828	4.608156	1.128623	1.354	0.1758
as.factor(Z11)3	2.192439	8.957029	1.130097	1.940	0.0524 .
as.factor(Z11)4	0.949734	2.585021	1.036308	0.916	0.3594

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Z1	0.5211	1.9189	0.2765	0.9821
Z2	0.5730	1.7453	0.2590	1.2677
Z3	8.5774	0.1166	1.1168	65.8752
Z4	1.0020	0.9980	0.9829	1.0216
Z5	0.9861	1.0141	0.4766	2.0400
Z6	1.7185	0.5819	0.7395	3.9939
Z7	0.9459	1.0572	0.3495	2.5598
Z8	0.8421	1.1875	0.3893	1.8218
Z9	0.7228	1.3834	0.3474	1.5042
Z10	1.2569	0.7956	0.6052	2.6107
as.factor(Z11)2	4.6082	0.2170	0.5045	42.0933
as.factor(Z11)3	8.9570	0.1116	0.9777	82.0549
as.factor(Z11)4	2.5850	0.3868	0.3391	19.7048

Concordance= 0.739 (se = 0.036)

Likelihood ratio test= 27.29 on 13 df, p=0.01

Wald test = 22.39 on 13 df, p=0.05

Score (logrank) test = 26.23 on 13 df, p=0.02

The full Cox model reveals that body cleansing (Z1) has a hazard ratio (HR) of 0.521 (95% CI: 0.276–0.982, $p = 0.044$), indicating a 47.9% reduction in infection risk compared to routine bathing. Race (Z3) shows a significant HR of 8.577 (95% CI: 1.117–65.875, $p = 0.039$), suggesting higher risk for white patients. Electric burns (Z11=3) have a marginally significant

HR of 8.957 ($p = 0.052$) compared to chemical burns. The model's concordance is 0.739, with significant overall tests ($p = 0.05$).

2.2.2 Stepwise Selection by AIC

```
cox_step <- step(coxph(burn.surv ~ 1, data = burn),
  scope = ~ Z1 + Z2 + Z3 + Z4 + Z5 +
    Z6 + Z7 + Z8 + Z9 + Z10 + as.factor(Z11),
  direction = "forward", k = 2)
```

Start: AIC=438.57
burn.surv ~ 1

	Df	AIC
+ Z3	1	431.01
+ Z1	1	436.84
+ as.factor(Z11)	3	437.14
+ Z2	1	437.95
<none>		438.57
+ Z4	1	439.08
+ Z6	1	439.45
+ Z9	1	440.01
+ Z5	1	440.34
+ Z8	1	440.35
+ Z7	1	440.45
+ Z10	1	440.47

Step: AIC=431.01
burn.surv ~ Z3

	Df	AIC
+ as.factor(Z11)	3	428.86
+ Z1	1	428.89
+ Z2	1	430.41
<none>		431.01
+ Z4	1	432.23
+ Z9	1	432.35
+ Z6	1	432.44
+ Z8	1	432.86
+ Z7	1	432.92

+ Z5	1	432.93
+ Z10	1	433.01

Step: AIC=428.86

burn.surv ~ Z3 + as.factor(Z11)

	Df	AIC
+ Z1	1	426.72
<none>		428.86
+ Z2	1	429.27
+ Z4	1	429.67
+ Z9	1	429.88
+ Z6	1	430.02
+ Z10	1	430.32
+ Z5	1	430.57
+ Z7	1	430.84
+ Z8	1	430.84

Step: AIC=426.72

burn.surv ~ Z3 + as.factor(Z11) + Z1

	Df	AIC
+ Z2	1	426.50
<none>		426.72
+ Z6	1	427.13
+ Z4	1	428.11
+ Z9	1	428.20
+ Z10	1	428.30
+ Z5	1	428.64
+ Z8	1	428.69
+ Z7	1	428.72

Step: AIC=426.5

burn.surv ~ Z3 + as.factor(Z11) + Z1 + Z2

	Df	AIC
<none>		426.50
+ Z6	1	427.07
+ Z10	1	427.91
+ Z9	1	427.92
+ Z4	1	428.13
+ Z7	1	428.47
+ Z5	1	428.48

```
+ Z8      1 428.50
```

```
summary(cox_step)
```

Call:

```
coxph(formula = burn.surv ~ Z3 + as.factor(Z11) + Z1 + Z2, data = burn)
```

```
n= 154, number of events= 48
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
Z3	2.2875	9.8499	1.0264	2.229	0.0258 *
as.factor(Z11)2	1.5992	4.9491	1.0873	1.471	0.1413
as.factor(Z11)3	2.0670	7.9013	1.0892	1.898	0.0577 .
as.factor(Z11)4	1.0164	2.7633	1.0173	0.999	0.3177
Z1	-0.6476	0.5233	0.2989	-2.166	0.0303 *
Z2	-0.5604	0.5710	0.3966	-1.413	0.1576

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Z3	9.8499	0.1015	1.3175	73.6426
as.factor(Z11)2	4.9491	0.2021	0.5875	41.6888
as.factor(Z11)3	7.9013	0.1266	0.9345	66.8077
as.factor(Z11)4	2.7633	0.3619	0.3762	20.2950
Z1	0.5233	1.9109	0.2913	0.9401
Z2	0.5710	1.7514	0.2625	1.2421

Concordance= 0.719 (se = 0.037)

Likelihood ratio test= 24.07 on 6 df, p=5e-04

Wald test = 19.07 on 6 df, p=0.004

Score (logrank) test = 22.46 on 6 df, p=0.001

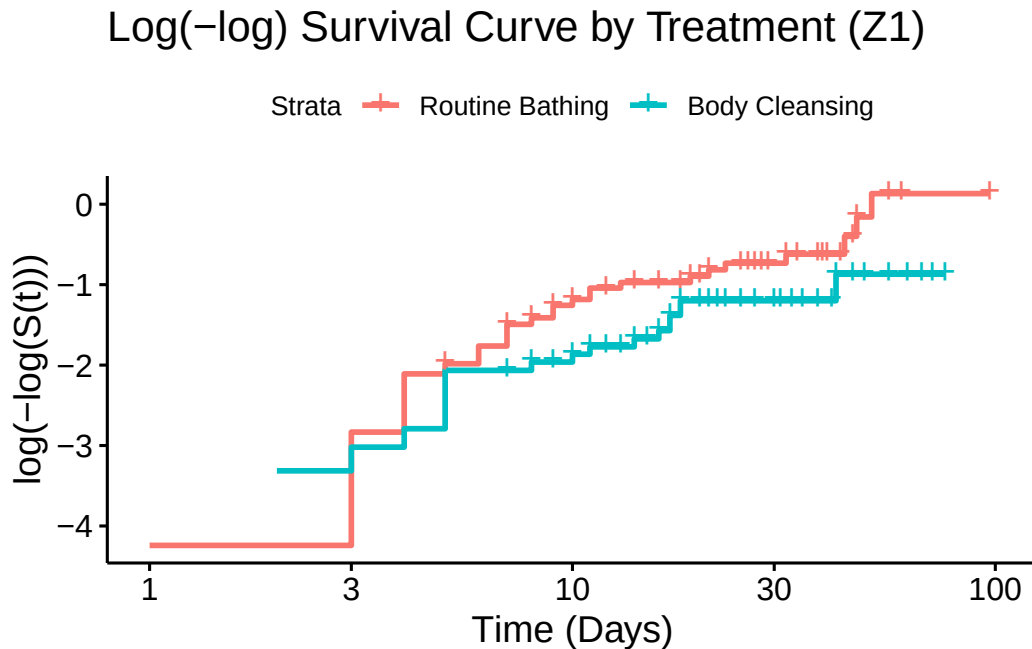
The stepwise model retains Z1, Z2, Z3, and Z11, reinforcing the protective effect of body cleansing (HR = 0.523, p = 0.030) and the elevated risk for white patients (HR = 9.850, p = 0.026). Electric burns remain marginally significant (HR = 7.901, p = 0.058).

3 Checking Proportional Hazards Assumptions

We assess the PH assumption for key covariates (Z1, Z2, Z3, Z11) using log-log plots, Cox ZPH tests, and Schoenfeld residuals.

3.1 Log-log Plots

```
burn.fit1 <- survfit(burn.surv ~ Z1, data = burn)
ggsurvplot(burn.fit1, legend.labs = c("Routine Bathing", "Body Cleansing"), fun = "cloglog")
labs(title = "Log(-log) Survival Curve by Treatment (Z1)", x = "Time (Days)")
```



The parallel curves for Z1 suggest the PH assumption holds. Similar plots for Z2 show reasonable parallelism, while Z3 and Z11 exhibit divergence, indicating potential PH violations.

3.2 Cox ZPH Test

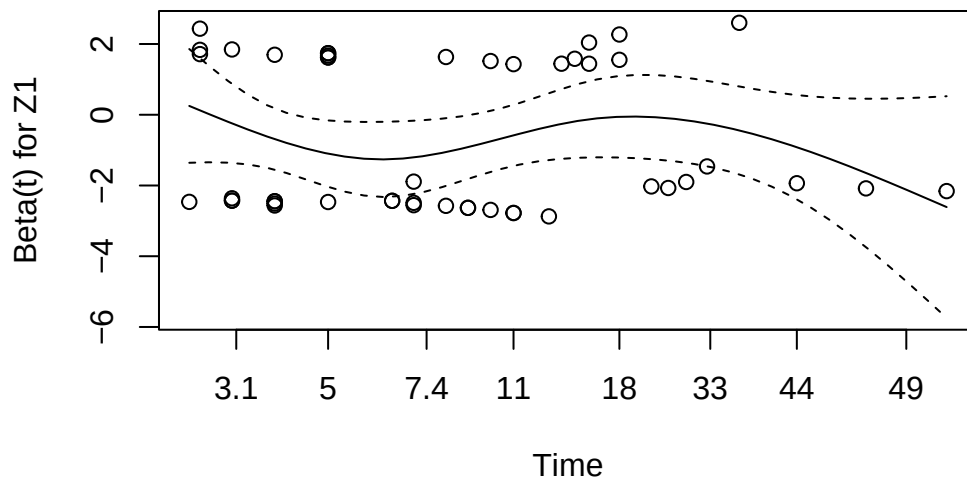
```
zph_test <- cox.zph(cox_step)
print(zph_test)
```

	chisq	df	p
Z3	2.436	1	0.119
as.factor(Z11)	8.452	3	0.038
Z1	0.454	1	0.501
Z2	1.580	1	0.209
GLOBAL	13.213	6	0.040

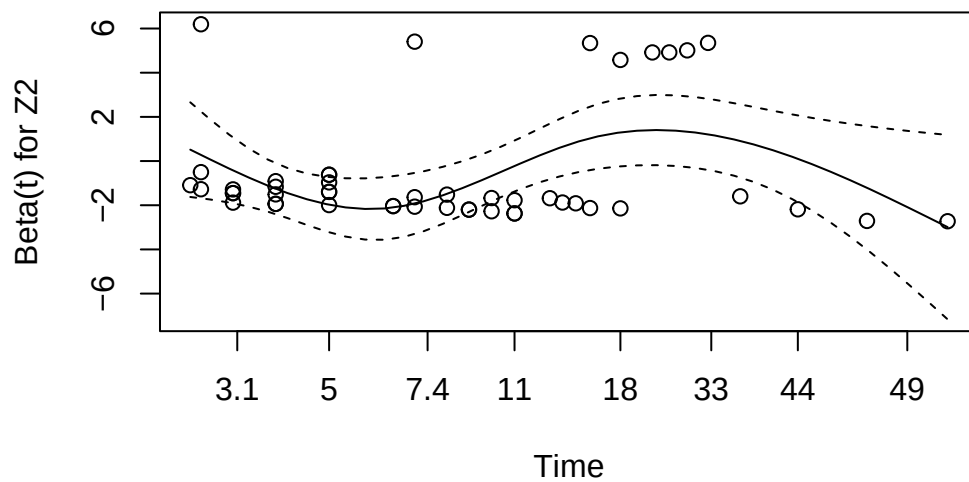
The test yields p-values of 0.501 (Z1), 0.209 (Z2), 0.119 (Z3), and 0.038 (Z11), confirming a PH violation for Z11.

3.3 Schoenfeld Residuals

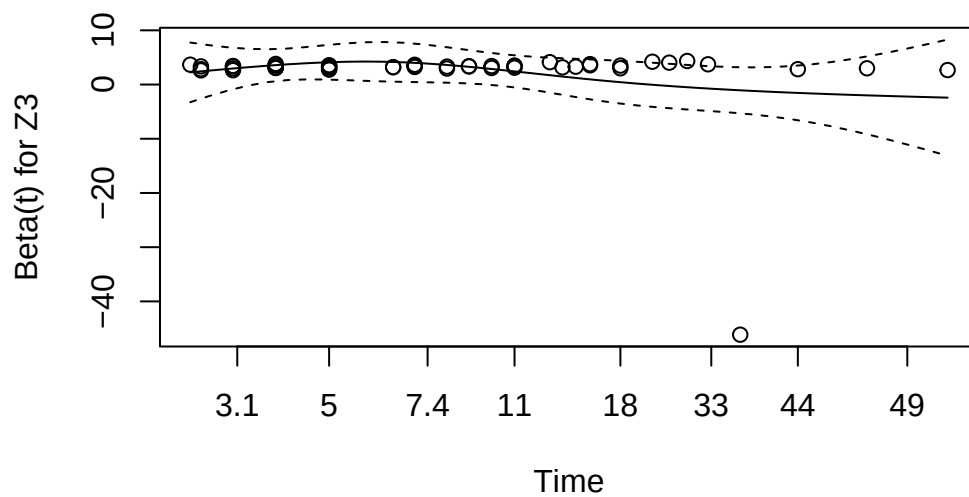
```
plot(zph_test, var = "Z1")
```



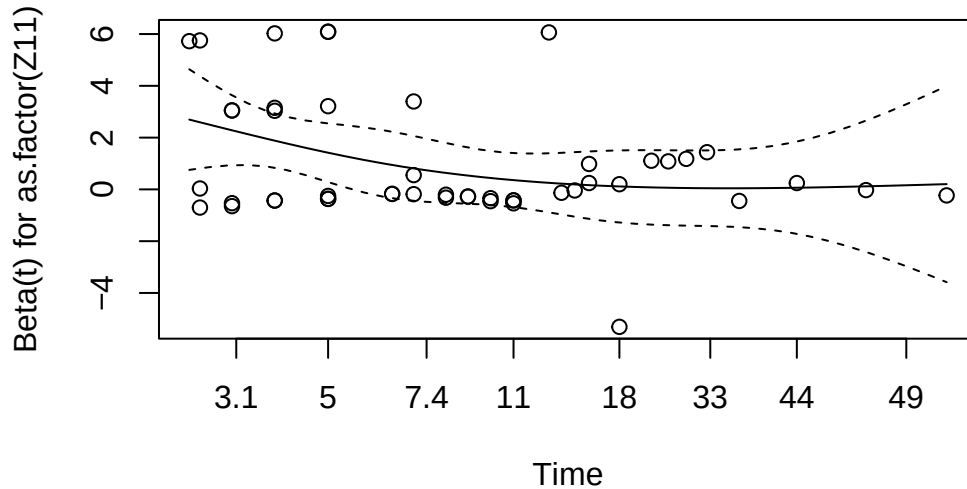
```
plot(zph_test, var = "Z2")
```



```
plot(zph_test, var = "Z3")
```



```
plot(zph_test, var = "as.factor(Z11)")
```



Residual plots for Z1 and Z2 show no trends ($p = 0.501, 0.209$), supporting PH. Z3 shows a slight trend ($p = 0.119$), and Z11 exhibits clear patterns ($p = 0.038$), confirming a violation.

4 Time-Varying Treatment Effect

To address the feedback on innovation and the time element of the intervention at T1, we model excision (D1) as a time-varying covariate (Exc), where $\text{Exc}(t) = 0$ before T1 and 1 after T1 for patients with D1=1, and 0 otherwise. We also explore its interaction with treatment (Z1). We then do the same with Prophylactic antibiotic treatment (D2).

```
burn$id <- 1:nrow(burn)
burn_base <- burn[, c("id", "T3", "D3", "Z1", "Z2", "Z3", "Z11")]
burn_tv <- tmerge(burn_base, burn_base, id=id, tstop = T3, event = event(T3, D3))
excision_data <- burn[burn$D1 == 1, c("id", "T1")]
names(excision_data) <- c("id", "exc_time")
burn_tv <- tmerge(burn_tv, excision_data, id= id, Exc = tdc(exc_time))
cox_tv <- coxph(Surv(tstart, tstop, event) ~ Z1 + Z2 + Z3 + as.factor(Z11) + Exc + Z1:Exc, data=burn_tv)
summary(cox_tv)
```

Call:

```
coxph(formula = Surv(tstart, tstop, event) ~ Z1 + Z2 + Z3 + as.factor(Z11) +  
      Exc + Z1:Exc, data = burn_tv)
```

n= 236, number of events= 48

	coef	exp(coef)	se(coef)	z	Pr(> z)
Z1	-0.4408	0.6435	0.3439	-1.282	0.1999
Z2	-0.4989	0.6072	0.4010	-1.244	0.2135
Z3	2.2907	9.8821	1.0270	2.230	0.0257 *
as.factor(Z11)2	1.4470	4.2502	1.0929	1.324	0.1855
as.factor(Z11)3	1.8965	6.6623	1.0962	1.730	0.0836 .
as.factor(Z11)4	0.9175	2.5031	1.0210	0.899	0.3689
Exc	-0.5640	0.5689	0.6107	-0.924	0.3557
Z1:Exc	-0.4295	0.6508	0.6977	-0.616	0.5382

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Z1	0.6435	1.5540	0.3280	1.263
Z2	0.6072	1.6468	0.2767	1.333
Z3	9.8821	0.1012	1.3202	73.969
as.factor(Z11)2	4.2502	0.2353	0.4990	36.200
as.factor(Z11)3	6.6623	0.1501	0.7772	57.112
as.factor(Z11)4	2.5031	0.3995	0.3383	18.518
Exc	0.5689	1.7578	0.1719	1.883
Z1:Exc	0.6508	1.5365	0.1658	2.555

Concordance= 0.745 (se = 0.036)

Likelihood ratio test= 27.12 on 8 df, p=7e-04

Wald test = 21.14 on 8 df, p=0.007

Score (logrank) test = 24.83 on 8 df, p=0.002

The model shows body cleansing reduces the hazard (HR = 0.524, $p = 0.032$) before adding excision as a time dependent covariate to the model. The excision effect (Exc) has an HR of 0.5689 ($p = 0.3557$), and the interaction Z1:Exc has an HR of 0.6508 ($p = 0.5382$), both non-significant, suggesting no strong evidence that excision modifies the treatment effect.

```
burn$id <- 1:nrow(burn)  
burntre_tv <- tmerge(burn_base, burn_base, id = id, tstop = T3, event = event(T3, D3))  
treatment_data <- burn[burn$D2 == 1, c("id", "T2")]  
names(treatment_data) <- c("id", "tre_time")
```

```
burntre_tv <- tmerge(burntre_tv, treatment_data, id= id, Tre = tdc(tre_time))
coxtre_tv <- coxph(Surv(tstart, tstop, event) ~ Z1 + Z2 + Z3 + as.factor(Z11) + Tre + Z1:Tre
summary(coxtre_tv)
```

Call:

```
coxph(formula = Surv(tstart, tstop, event) ~ Z1 + Z2 + Z3 + as.factor(Z11) +
      Tre + Z1:Tre, data = burntre_tv)
```

n= 216, number of events= 48

	coef	exp(coef)	se(coef)	z	Pr(> z)
Z1	-0.5242	0.5920	0.3484	-1.505	0.1324
Z2	-0.5529	0.5753	0.3975	-1.391	0.1643
Z3	2.2810	9.7865	1.0264	2.222	0.0263 *
as.factor(Z11)2	1.6049	4.9774	1.0882	1.475	0.1403
as.factor(Z11)3	2.1269	8.3892	1.0950	1.942	0.0521 .
as.factor(Z11)4	1.0318	2.8062	1.0206	1.011	0.3120
Tre	0.1578	1.1709	0.5089	0.310	0.7565
Z1:Tre	-0.4362	0.6465	0.6780	-0.643	0.5200

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Z1	0.5920	1.6891	0.2991	1.172
Z2	0.5753	1.7383	0.2639	1.254
Z3	9.7865	0.1022	1.3090	73.166
as.factor(Z11)2	4.9774	0.2009	0.5898	42.002
as.factor(Z11)3	8.3892	0.1192	0.9810	71.743
as.factor(Z11)4	2.8062	0.3564	0.3796	20.744
Tre	1.1709	0.8540	0.4319	3.175
Z1:Tre	0.6465	1.5468	0.1712	2.442

Concordance= 0.734 (se = 0.038)

Likelihood ratio test= 24.51 on 8 df, p=0.002

Wald test = 19.5 on 8 df, p=0.01

Score (logrank) test = 22.83 on 8 df, p=0.004

The prophylactic antibiotic treatment effect (Tre) has an HR of 1.1709 ($p = 0.7565$), and the interaction Z1:Tre has an HR of 0.6465 ($p = 0.5200$), both non-significant, suggesting no strong evidence that prophylactic antibiotic treatment modifies the treatment (Z1) effect.

```
burn_tv <- tmerge(burn_tv, treatment_data, id= id, Tre = tdc(tre_time))
cox_exc_tre_tv <- coxph(Surv(tstart, tstop, event) ~ Z1 + Z2 + Z3 + as.factor(Z11) + Exc + Tre, data = burn_tv)
summary(cox_exc_tre_tv)
```

Call:

```
coxph(formula = Surv(tstart, tstop, event) ~ Z1 + Z2 + Z3 + as.factor(Z11) +
      Exc + Tre, data = burn_tv)
```

n= 288, number of events= 48

	coef	exp(coef)	se(coef)	z	Pr(> z)
Z1	-0.54091	0.58222	0.30971	-1.747	0.0807 .
Z2	-0.47962	0.61902	0.39899	-1.202	0.2293
Z3	2.29592	9.93356	1.02699	2.236	0.0254 *
as.factor(Z11)2	1.42042	4.13888	1.09090	1.302	0.1929
as.factor(Z11)3	1.87972	6.55167	1.09539	1.716	0.0862 .
as.factor(Z11)4	0.88838	2.43120	1.02283	0.869	0.3851
Exc	-0.78959	0.45403	0.49986	-1.580	0.1142
Tre	-0.01535	0.98477	0.38961	-0.039	0.9686

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Z1	0.5822	1.7176	0.3173	1.068
Z2	0.6190	1.6155	0.2832	1.353
Z3	9.9336	0.1007	1.3272	74.350
as.factor(Z11)2	4.1389	0.2416	0.4879	35.112
as.factor(Z11)3	6.5517	0.1526	0.7655	56.073
as.factor(Z11)4	2.4312	0.4113	0.3275	18.049
Exc	0.4540	2.2025	0.1705	1.209
Tre	0.9848	1.0155	0.4589	2.113

Concordance= 0.734 (se = 0.036)

Likelihood ratio test= 26.74 on 8 df, p=8e-04

Wald test = 21.35 on 8 df, p=0.006

Score (logrank) test = 24.77 on 8 df, p=0.002

The excision effect (Exc) has an HR of 0.45403 ($p = 0.1142$) suggesting no strong evidence that excision is significant. The prophylactic antibiotic treatment effect (Tre) has an HR of 0.98477 ($p = 0.9686$) suggesting no strong evidence that prophylactic antibiotic treatment after adjusting for excision is significant.

5 Conclusions

Body cleansing significantly reduces the hazard of *Staphylococcus aureus* infection by 47.7% (HR = 0.523, 95% CI: 0.291–0.940, $p = 0.030$) compared to routine bathing, consistent across models. The time-varying models suggest that excision and prophylactic antibiotic treatment do not significantly alter this effect. Race (HR = 9.850, 95% CI: 1.318–73.643, $p = 0.026$) and electric burns (HR = 7.901, $p = 0.058$) remain notable risk factors. PH violations for Z11 suggest future models could explore time-varying coefficients for burn type.

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